

Active Inference in Rust.... For Ants? (4-4-2024 Active InferAnts bundle of

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all right it's April 4th 2024 this is an update from the active infer ants project where we're applying active inference models to ants in 2021 working with other authors we published an initial python code model ant simulation with Niche modification and forging with the single pheromone code was written initially by Alec in that case adapting it from a previous publication with axle at all that was looking at cultural patterns of iates and with other nestmates in the subsequent years we continued to work on and explore these different ideas related to social physiology cyber physical systems multiscale planning and behavior and ant colonies and so on and today I wanted to bundle together a couple of updates related to the project as there's just a great Confluence of factors ranging from the RX and fur team's live stream on building a drone earlier today to just the inspiring work in the RX andur project at the instit and Holly's encouragement to look into rust so the big announcement SLT takeaway over at active inference Institute GitHub slactive infer ant let's start from scratch and multiple places and converge and diverge and have fun with ants and active inference and for the rest of this video I'm going to start zoomed in in the first section talking a little bit methodologically

about how I used cursor.sh to translate RX and F Julia code and pmdp python code into new rust code and publish that open source using a method with cursor that pulls in GitHub repos and then we'll zoom out a little bit and look at how this draft SL sketch of the active inference package is laid out and what was retained through iterating locally on my side and as mentioned before I expect that we'll start from multiple places and multiple step by steps so I encourage that people get involved this fun project and today I'll just show a little bit that might spark some thoughts and uh let's just see how to also use these tools in a cool way so I'm starting in folder _other with the Clone GitHub users python script this script does work not all the scripts work and the payload of this script is that for whichever GitHub usernames are provided it downloads all of the open repos that that user has into a folder called repo so here I've only left selected bias lab which is the developers of RX andur Julia and infer actively which is the hosting group of pmdp however I also curated some lists of a bunch of other active inference usernames and also some other things that are fun to bring in like bringing in energetics and quadres tetrahedral coordination geometries as well as a lot of ants and bioinformatics and also the exciting development in multi-agent Frameworks many could be added to all these lists but just to kind of show the

minimal example what the current status of the repo does is it clones in the repos of bias lab and infer actively and puts it into other then by going to the gear on the top right of cursor and then features resync index cursor re indexes all of the code which helps it for when you do the AI chat you can do control enter and chat with the code base so that is a super helpful feature as is selecting text and control K or doing control shift L to move something over into the side window so all these are methods that I'm sure other people also have things that are exciting for them and there's probably many more improvements and meta improvements to come but this clone GitHub user script allowed me to bring in exactly the genres and the repos one by one in the repo script or in a batch download with the users and then cursor enabled me to chat with the codebase to explain it better and also to ask now write a pmdp styled script for this situation so um two ways in which I use that first I'll go through the rust read me and then some of the rust translations from the PDP and RX and fer and then we'll return to how the ants looked in the second section so this is in the systems design section of the repo because it's very preliminary the goals of today's work essentially where to use language models and code and document methods to translate and transform some open source Python and Julia active inference code

into open source rust active inference
code to develop a noisy non-expert draft
sketch of an active inference ant colony
boldy scale simulation in Rust language
in order to learn highlight and explore
the unique performance and security
features of rust language for beginners
such as myself and to provide an initial
basis for further open source
development at the intersection of
active inference rust and ants so I used
cursor

0.331 with the GPT form model I use the
Clone GitHub user script to bring in as
many repos as possible related to active
inference or just a few resync the index
in cursor copied over from the rest of
the script some of the Python files and
renamed them to be rust files and then
here's some of the prompts that I used
in the side chat and in the control K
chat popup to help transform and iterate
on improve the code so these were just a
few of the prompts that

helped where it got me was some Rust
looking PDP style ants that I copied
over and multiple separate attempts to
write rust looking code not all of it
was tested and a lot of it is still
schematic but it's outlining at least an
approach that with a rust expert in the
loop I think we could have a great time
with the generalized multiscale modeling
of

things thing

class and this rust things starts in the
kind of agent-based approach along the
lines of the AOS active entity ontology

for Science and then I also tried
several other features based upon the
code models know of rust so rust meta
has some simulation
environment aspects rust system check
prints a lot of information about the
computer rust meta
configs has different kinds of
affordances for the then configuration
file which sets a bunch of the
parameters which are used later and then
in the rust worlds folder under config
this is even more speculative
so here we have cognitive security
classes in

Rust security classes in Rust and
porting over from 4D Solutions quadres
tetrahedral coordinates and Tetra
volumes those were both translated from
python so this just shows that by
bringing in repos related to active
inference we can transfer the semantics
of examples and hopefully there will be
a growing library of open- source
examples in all these different
languages and transfer and interpolate
within and outside of those bounds to
translate code and to get somewhere fast
with natural conversation which is super
exciting so now pulling back a little
bit folder zero in the
repo has some system specifications that
were also iteratively written with
language models and my put and then the
folders are as follows nine other has
some scripts related to cloning in repos
six API is for API calls which is also
just a kind of broadly outlined set of

methods most of the simulating and preparing and all of the operating is in the scripts 1 through five so in one there's prepare so this has all of the agent-based definitions including pmdp ants that are defined and the general and the config set different configs most of these variables are not called however it's fun to explore and just build these hypothetical meta configuration files and then in the general preparation we have some scripts pointing towards aspects of cognitive security the course ordiates the IVM and the XYZ coordinates being able to flip between the coordination systems implementing some cryptographic metaphysical systems nesting networking Network science different physics including the metaphysics and the cognitive physics and then also for a little comedic relief the Salt Fat acid heat simulator in pmdp format so prepare gets all of the configurations ready for assembling the different things operate has a plan render and execute three-step process with a situational ant awareness Observer role that actually prepares plans out and then executes computationally the simulation measure has a set of scripts and methods related to specifying how you want to measure in a Quantum cognitive frame and then how you're going to summarize categorize and apply statistics to the measurements report is currently just

for this release the specify report
methods that detail how reporting is
going to occur and then the
specifications for followup which are
the implications of the report so that's
just one top level folder scheme for
multi-layer simulations like the active
inference package has to be and I look
forward to hearing about some radically
alternative starting points that we can
also have in our Colony so hope this is
at least interesting to some of you I
encourage people with some or a lot or
in between amount of familiarity with
active inference and ants and all these
different topics to come get involved
and come to our weekly RX and fur
meetings on Thursday join The Institute
Discord or email blanket active
inference do Institute or get in touch
another way because this is going to be
a fun project and really looking forward
to seeing how we can all start over and
collaborate on this further all
right